08/2/2024 FN

Hall Ticket Number:

MR-22

Code: B324CB2

MAHATMA GANDHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.Tech. III Semester Regular Examinations, Jan/Feb-2024

Discrete Mathematics

(Common to CSB & CSD)

Time: 3 Hours | ***Answer ALL Questions*** | Max. Marks: 60

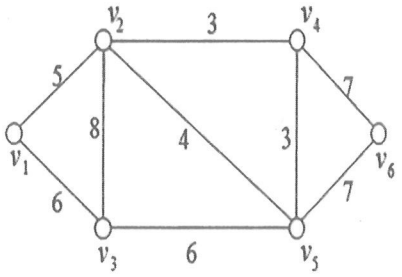
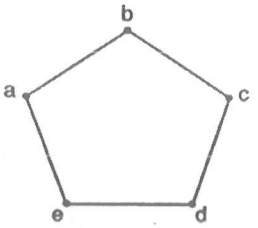
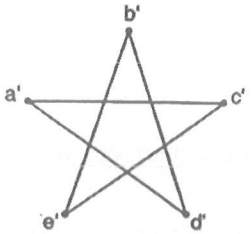
Part-A (10 x 1 = 10 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
1. a)	Define tautology.	1	1	1	1
1. b)	Construct truth table for $P \wedge Q$.	1	2	1	1
1. c)	What is the Cartesian product of A = {1,2} and B = {a, b, c}?	1	1	2	1
1. d)	Find the inverse of the function $f(x) = x^3 + 1$.	1	2	2	1
1. e)	Define a monoid.	1	1	3	1
1. f)	Define a group.	1	1	3	1
1. g)	Define combinations.	1	1	4	1
1. h)	Find the relation between nC_r and $(n + 1)C_r$.	1	2	4	1
1. i)	Define Chromatic Number.	1	1	5	1
1. j)	Define Planar Graph.	1	1	5	1

Part-B (5 x 10 = 50 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
2. a)	Obtain PDNF of $(p \wedge q) \vee (\sim p \wedge r) \vee (q \wedge r)$ using truth table.	5	2	1	1
2. b)	Show that $R \wedge (P \vee Q)$ is a valid conclusion from the premises $P \vee Q$, $Q \rightarrow R$, $P \rightarrow M$ and M.	5	3	1	1
	OR				
2. c)	Show that $(\forall x)\{P(x) \vee Q(x)\} \Rightarrow (\forall x) P(x) \vee (\exists x)Q(x)$.	5	2	1	1
2. d)	Explain in detail about logical connectives with examples.	5	2	1	1
3. a)	Find the transitive closure of the relation R= {(1,2), (2,3), (3,4), (4,1)}.	5	2	2	1
3. b)	Let X= {1,2,3,4,5,6} and R= {(1,4), (1,3), (1,2), (2,5), (2,4), (2,3), (3,4), (3,6), (5,6)}. Define a compatibility relation. Draw the graph of the relation R.	5	3	2	1
	OR				
3. c)	Let A={1, 2, 3, 4, 6}, and let R be the relation on A defined by "x divides y", written x|y. (Note: x|y iff there exists an integer z such that xz = y) i) Write R as a set of ordered pairs. ii) Draw the digraph of R. iii)Draw the Hasse diagram of the relation	5	3	2	1
3. d)	Let A = {1,2,3,4}, B = {a, b, c}and C = {w, x, y, z}with f: A → B and g: B → C given by f = {(1, a), (2, a), (3, b), (4, c)}, g = {(a, x), (b, y), (c, z)}find gof.	5	2	2	1
4. a)	Prove that the set of even integers forms a group under addition.	5	3	3	1
4. b)	Prove that the set{1,5,7,11}is a finite abelian group of order 4 under multiplication modulo 12 as composition.	5	3	3	1
	OR				
4. c)	On the set Q of all rational numbers, the operation * is defined by a*b=a+b-ab. Verify that under this operation, Q forms a group or not?	5	2	3	1
4. d)	Let G = {1, −1, i, −i}be a multiplicative group. Find the order of every element.	5	2	3	1

P.T.O.

5. a)	Define binomial theorem. What is the coefficient of $x^{101}y^{99}$ in the expansion of $(2x-3y)^{200}$?	5	3	4	1
5. b)	In how many ways 4 boys and 4 girls can be seated in a row so that boys and girls are alternate?	5	3	4	1
	OR				
5. c)	There are 7 men and 3 women. Find the number of ways in which a committee of 6 persons can be formed if the committee is to have (i) exactly 4 men, and (ii) at least 2 women.	5	3	4	1
5. d)	Find the number of integers solutions of $x_1 + x_2 + x_3 + x_4 + x_5 = 30$. Where $x_1 \geq 2, x_2 \geq 3, x_3 \geq 4, x_4 \geq 2, x_5 \geq 0$.	5	2	4	1
6. a)	Prove that a complete graph K_n is planar if and only if $n \leq 4$.	5	3	5	1
6. b)	Explain kruskal's algorithm and using the same obtain the minimal spanning tree for the following weighted graph. 	5	2	5	1
	OR				
6. c)	Show that the given pair of graphs are isomorphic. G:  H: 	5	2	5	1
6. d)	Differentiate between Eulerian graph and Hamiltonian graph with an example each.	5	2	5	1

M: Marks; L: Bloom's Taxonomy Level; CO: Course Outcome; PO: Programme Outcome